Math 540
Homework #1

Name: Gianluca Crescenzo

Exercise 1 (Section 12 & 13, Problem 2)**.** *Consider the nine topologies on the set* $X = \{a, b, c\}$. *Compare them, that is, for each pair of topologies, determine whether they are comparable, and if so, which is the finer.*

Proof. □

Exercise 2 (Section 12 & 13, Problem 3)**.** *Show that the collection*

$$T_c := \{U \mid X - U \text{ is countable or } X - U = X\}$$

is a topology on X. *Is the collection* $T_\infty := \{U \mid X - U$ *is infinite or empty or all of* $X\}$ *a topology on* X*?*

Proof. Clearly $\emptyset, X \in T_c$. Let $\{U_i\}_{i \in I} \subseteq X$ be an arbitrary family of open sets in X. Then $(\bigcup_{i\in I} U_i)^c = \bigcap_{i \in I} U_i^c$. Since each U_i^c is countable, the set $(\bigcup_{i \in I} U_i)^c$ will be the empty set or at most a countable set. Now let $\{U_i\}_{i=1}^n \subseteq X$ be a finite collection of open sets in X. Then $\left(\bigcap_{i=1}^n U_i\right)^c = \bigcup_{i=1}^n U_i^c$. Since finite unions of countable sets are countable, the set $\left(\bigcap_{i=1}^n U_i\right)^c$ is countable. Thus T_c is a topology on X.

The set T_∞ does not necessarily form a topology on X. As a counter example, take $X = [0, 1]$. Consider the family of sets $\left\{\left(\frac{1}{n}, 1\right]\right\}_{n=1}^\infty$. Note that $\left(\frac{1}{n}, 1\right] \in T_\infty$ for all n because $\operatorname{card}\left(\left(\frac{1}{n}, 1\right]\right) = \operatorname{card}\left(\left[0, \frac{1}{n}\right]\right) = \infty$. But:

$$\begin{aligned}
\left(\bigcup_{n=1}^\infty (\frac{1}{n}, 1]\right)^c &= \bigcap_{n=1}^\infty (\frac{1}{n}, 1]^c \\
&= \bigcap_{n=1}^\infty \left[0, \frac{1}{n}\right]^c \\
&= \{0\}.
\end{aligned}$$

Hence $\bigcup_{n=1}^\infty \left(\frac{1}{n}, 1\right] \notin T_\infty$. Since there exists a collection of open sets whose union is not open, T_∞ is not a topology on $[0, 1]$. □

Exercise 3.

Proof. □

Exercise 4 (Section 15 & 16, Problem 1)**.** *Show that if* Y *is a subspace of* X *and* A *is a subset of* Y, *then the topology* A *inherits as a subspace of* Y *is the same as the topology it inherits as a subspace of* X.

Proof. Let (X, T) be a topological space. Let $T_Y := \{Y \cap U \mid U \in T\}$. The topology A inherits as a subspace of Y is:

$$\begin{aligned}\{A \cap V \mid V \in T_y\} &= \{A \cap (Y \cap U) \mid U \in T\} \\ &= \{A \cap U \mid U \in T\},\end{aligned}$$

which is precisely the topology A inherits as a subspace of X. □

Exercise 5 (Section 12 & 13, Problem 2)**.** *If* T *and* T' *are topologies on* X *and* T' *is strictly finer than* T*, what can you say about the corresponding subspace topologies on the subset* Y *of* X*?*

Proof. Since T' is strictly finer than T, this means $T \subset T'$. Let:

$$\begin{aligned}T_Y &:= \{Y \cap U \mid U \in T\}, \\ T'_Y &:= \{Y \cap U \mid U \in T'\}.\end{aligned}$$

Clearly $T_Y \subset T'_Y$, hence T'_Y is strictly finer than T_Y. □

Exercise 6 (Section 15 & 16, Problem 3)**.** *Consider the set* $Y = [-1, 1]$ *as a subspace of* $\mathbb{R}$*. Which of the following sets are open in* Y*? Which are open in* $\mathbb{R}$*?*

$$\begin{aligned}A &:= \left\{x \mid \frac{1}{2} < |x| < 1\right\}, \\ B &:= \left\{x \mid \frac{1}{2} < |x| \leqslant 1\right\}, \\ C &:= \left\{x \mid \frac{1}{2} \leqslant |x| < 1\right\}, \\ D &:= \left\{x \mid \frac{1}{2} \leqslant |x| <\leqslant 1\right\}, \\ E &:= \left\{0 < |x| < 1 \textit{ and } \frac{1}{x} \notin \mathbb{Z}_+\right\}.\end{aligned}$$

Proof. The set A is open in Y and $\mathbb{R}$. The set B is open in Y but not open in $\mathbb{R}$. The sets C and D are neither open in Y nor open in $\mathbb{R}$. For E, note that it can be realized as the set:

$$E = \bigcup_{n=1}^{\infty} \left(-\frac{1}{n}, -\frac{1}{n+1}\right) \cup \bigcup_{n=1}^{\infty} \left(\frac{1}{n+1}, -\frac{1}{n}\right).$$

The sets $\left(-\frac{1}{n}, -\frac{1}{n+1}\right)$ and $\left(\frac{1}{n+1}, -\frac{1}{n}\right)$ are open in Y and open in $\mathbb{R}$. Since an arbitrary union of open sets is open, E is surely open in Y and open in $\mathbb{R}$. □

Exercise 7 (Section 15 & 16, Problem 4)**.** *A map* $f : X \to Y$ *is said to be an open map if for every open set* U *of* X*, the set* $f(U)$ *is open in* Y*. Show that* $\pi_1 : X \times Y \to X$ *and* $\pi_2 : X \times Y \to Y$ *are open maps.*

Proof. Let $\mathcal{B} = \{U \times V \mid U \subseteq X \text{ open }, V \subseteq X \text{ open }\}$. This basis generates the product topology on $X \times Y$. Let $\mathcal{O} \subseteq X \times Y$ be open. Let $(x, y) \in \mathcal{O}$. Then there exists a basis element $U_x \times V_y \in \mathcal{B}$ such that $(x, y) \in U \times V \subseteq \mathcal{O}$. We can write:

$$\mathcal{O} = \bigcup_{(x,y) \in \mathcal{O}} U_x \times V_y.$$

Since functions distribute over unions, $\pi_1(\mathcal{O})$ and $\pi_2(\mathcal{O})$ will be arbitrary unions of open sets, hence they are open. $\square$